

Unit 12, Lesson 6: Rewriting Expressions with Common Monomial Factors

Benchmark

MA.B.AR.1.3 Rewrite the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

Learning Target

- ☐ **I can rewrite** the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.

12.6.1 Warm-Up: Matching Solutions

1. Consider the unfinished equation $12(x - 3) + 18 = \underline{\hspace{2cm}}$.
- a. Match the following expressions with the number of solutions the equation would have if that expression were on the right side of the equation.

Expression
$6(2x - 3)$
$4(3x - 3)$
$4(2x - 3)$

Number of Solutions to the Equation
one solution
no solutions
infinitely many solutions

- b. Explain how you determined which expression would result in an equation with infinitely many solutions.

12.6.2 Guided Instruction: Factoring Monomials

While the area model is helpful for visually representing the factors of an expression, it is not the only strategy.

Area models normally are used to represent multiplication but can be used in the reverse to find the factors of an expression. Similarly, the distributive property is normally used to multiply expressions but the reverse, division, can be used to find the factors multiplied.

1. Consider the expression $6x^5 + 2x^2$.
 - a. What is the greatest common factor of $6x^5$ and $2x^2$?

Recall the quotient of powers law of exponents.

$$\frac{a^m}{a^n} = a^{m-n}$$

- b. Divide both terms in the expression by the **GCF** determined in Part A.

$$\frac{6x^5}{\boxed{}} = \boxed{}$$

$$\frac{2x^2}{\boxed{}} = \boxed{}$$

- c. Rewrite the expression using the distributive property.

$$\boxed{} (\boxed{} + \boxed{})$$

d. Verify using the distributive property that the expression you determined in Part C is equivalent to $6x^5 + 2x^2$.

e. Choose two different common factors to rewrite $6x^5 + 2x^2$.

____(____ + ____)

____(____ + ____)

f. Confirm using the distributive property that the expressions are equivalent.

2. Consider the expression $9g^4h^5 - 12gh^2$.

a. What is the GCF of the two terms in the expression?

b. Rewrite $9g^4h^5 - 12gh^2$ using the GCF as one of the factors.

c. Verify using the distributive property that the expression you determined in Part B is equivalent to the original expression.

12.6.3 Collaboration: Which is Equivalent?

Work with your partner to complete the following.

1. Only says the expressions $\frac{1}{2}x^2(8x + 10x^6)$ and $x^4(5x^5 + 4)$ are equivalent.

a. Do you agree with Only?

b. Explain why or why not.

2. Zeke and Ford each rewrote $3w^2x^2(3x^2 + 3w^6)$. Their expressions are shown.

Zeke's Expression	Ford's Expression
$9w^2x^4 + 3w^6$	$9w^2x^4 + 9w^8x^2$

- a. Who wrote an expression equivalent to $3w^2x^2(3x^2 + 3w^6)$?

- ☐ Zeke
- ☐ Ford
- ☐ Both Zeke and Ford

- b. If one of the students wrote an expression that was not equivalent, write a note explaining how to avoid the same mistake in the future. If both students were correct, explain how they can both be correct.

3. Lucía and Faye each rewrote $-21y^3 - 9y^4$ using a common monomial factor.

Lucía's Expression	Faye's Expression
$-3y^3(7 + 3y)$	$3y^3(7 - 3y)$

- a. Who wrote an expression equivalent to $-21y^3 - 9y^4$?
- ☐ Lucía
 - ☐ Faye
 - ☐ Both Lucía and Faye
- b. If one of the students wrote an expression that was not equivalent, write a note explaining how to avoid the same mistake in the future. If both students were correct, explain how they can both be correct.

12.6.4 Activity: Matching Expressions

Six expressions are provided in the table.

Work with your partner to find two expressions from the card set that are equivalent to each given expression. Write the letter of each card in the columns labeled Card 1 and Card 2. Then, create an additional equivalent expression not included on either card in the last column.

Given Expression	Equivalent Card 1	Equivalent Card 2	Equivalent Expression
$14q^5r^2 - 28q^6r^4$			
$9(-q^3r + 2q^2r^5)$			
$\frac{2}{3}(9qr^3 + 18q^5r^2)$			
$-q^7r^6 - 6q^4r^4$			
$2(10q^4r^7 + 20q^2r^3)$			
$-12q^3r + 15q^2r^5$			

12.6.5 Your Turn!

1. Complete the table.

Expression	Factored Expression
$16t^6 + 20t^4$	
	$3k^2(-2k + 5)$
$21m^4n - 27mn^3$	

12.6.6 Wrap-Up: Reflection

1. Rate your understanding of today's learning target on the emoji scale.

- ☐ **I can rewrite** the sum of two algebraic expressions having a common monomial factor as a common factor multiplied by the sum of two algebraic expressions.



